

Aliah University

End-Semester Examination (Autumn Semester) - 2023

5th Semester B.Tech (CSE))

Paper Name: Database Management System

Paper Code: CSEUGPC11

Full Marks: 80

Time: 3 hrs

Group-A

[1x5]

1. Select the best alternative for the following
 - a) $A \cap B$ is same as: i) $(A-B)-B$ ii) $(A \cup B)-A$ iii) $A-(A-B)$ iv) $A \cup (B-A)$
 - b) Grant statement is an example of: i) DDL ii) DML iii) DCL iv) DQL
 - c) Rollback ensures: i) Atomicity ii) Durability iii) Consistency iv) Isolation
 - d) Which of the following is not an aggregate function in SQL: i) SUM ii) MAX iii) COUNT iv) MEAN
 - e) After successful execution of the last instruction of the transaction, the state is i) Partially committed ii) Committed iii) Ready of abort iv) Aborted

Group- B

Answer any three

[5x3]

2. What are Amstrong's axioms? Prove the sixth inference rule.
3. Explain the ACID properties of a Transaction.
4. What is cascading rollback? What is an irrecoverable schedule?
5. Explain with examples - 2NF and 3NF.

Group-C

Answer any four

[15x4]

6. Considering the tables EMP(empid, name, sal, did) and DEPT(did, dname, dloc), write relational algebra and SQL for each of the query-
a) Who earns the maximum salary in the 'sales' department?
b) Which department has the minimum number of employees?
c) List the employee who earns more than the average salary of his/her department?
[4+3+8]
7. a) Define conflict and view serializability.
b) Justify with an example – "Every conflict serializable is viewed as serializable but not vice-versa".
c) Draw an ER Diagram for a university student registration system that includes teacher, students, courses, and enrollment. One teacher can offer multiple courses, but one course has only one teacher. A student can take one or more courses. Clearly indicate primary keys, foreign keys, attributes, and cardinality of relationships.
[5+5+5]
9. a) What is a two-phase locking protocol? What are its advantages?
b) Explain the algorithm of time-stamp-based locking protocol.
c) Describe a preemptive and non-preemptive scheme of deadlock avoidance.
[5+5+5]
10. a) Write the algorithm for finding the primary key and find out the candidate key(s) for the following – $R(A,B,C,D,E,F,G,H)$ and $F = \{ A \twoheadrightarrow CD, AB \twoheadrightarrow EG, C \twoheadrightarrow G, D \twoheadrightarrow EF, AC \twoheadrightarrow D, G \twoheadrightarrow FH \}$
b) Write the algorithm for finding the canonical cover of a FD set. Find the canonical cover of the following $F = \{ W \rightarrow X, Y \rightarrow X, Z \rightarrow WXY, WY \rightarrow Z \}$ [8+7]
[5+5+5]
11. a) Explain dense and sparse indexing.
b) Discuss dynamic hashing technique.
c) Explain the steps of query optimization.

Pessimistic